

STEPCLIMBER · REQUIREMENTS

StepClimber requirements

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	SCM-REQ-001	Version	0.4
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/stepclimber		
Source	docs/03-requirements.md @ 8763e13		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); R3, R5, R7 and R12 revised; status against SCM-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	Record Amish's 2026-09-25 decisions (R1 rating, R11 relaxed to 1,500 mm, R12 budget kept); status against SCM-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-24	Amish Chadha	Initial scaffold

StepClimber requirements

These requirements are checked by calculation in SCM-CAL-001 v0.2, which models the cluster and drive rework Amish decided on 2026-09-25 (SCM-DDR-002). On paper, ten are met, one is not met (R6) and four cannot be verified without hardware. Targets are still to be revised after user research (see SCM-PRB-001). Decisions are recorded in SCM-DDR-001 and SCM-DDR-002.

The **design load case** is a 60 kg (132 lb) payload on a 26.3 kg truck, 86.3 kg in total, climbing a residential stair with 196 mm (7.75 in) risers and 254 mm (10 in) treads, the steepest the US residential code allows.

Table 1. Requirements and status at TRL 3.

ID	Requirement	Target	Status at TRL 3 (SCM-CAL-001 v0.2)	Verification (later)
R1	Rated stair load	60 kg (132 lb) payload up and down stairs; 100 kg on the flat. Rating decided by Amish, 2026-09-25	Met: safety factors 1.8 (shaft), 1.7 (spider), 2.8 (rail) at a 3 g dropped-step load; fatigue not assessed	Proof load
R2	Stair range	Straight flights with risers 100 to 200 mm, treads 250 mm or more and nosing overhang up to 32 mm	Met: lands 66 mm past a square nosing at a 200 mm riser (climbs to 225 mm), needs 240 mm of tread at most, and the arms clear 32 mm overhangs	Stair survey; trial climbs
R3	Climb speed	17 steps/min or more, up and down, at rated load. Revised from 20 steps/min by Amish, 2026-09-25 (SCM-DDR-002)	Met: 238 W peak motor output on a 250 W motor (5 % margin)	Timed climb
R4	Endurance	1,000 or more loaded steps up plus 1,000 down per charge	Met: 1,327	Logged use
R5	Truck mass	27 kg or less with battery; battery removable and 3 kg or less. Revised from 25 kg by Amish, 2026-09-25 (SCM-DDR-002)	Met: 26.3 kg, pack 2.6 kg (0.7 kg margin)	Weighing
R6	Operator handle force	100 N or less at the grip while climbing inside the tilt window (R7)	Not met: 111 N at the plus or minus 6 degree window edge; 70 N at the set angle. Options in SCM-DDR-002 item 14, proposed, awaiting Amish	Force gauge
R7	Tilt control	IMU measures frame angle at 100 Hz or faster; the controller stores the balance angle when a climb starts, shapes motor speed to keep the angle within plus or minus 6 degrees of it, and stops and alerts within 0.2 s if the angle leaves that window or the range 15 to 45 degrees back from vertical. Window tightened from 8 degrees by Amish, 2026-09-25 (SCM-DDR-002)	Not verifiable at TRL 3 (control concept only)	Bench test
R8	Hold on any loss	Load holds on the stair with power off, grip released, battery removed or any fault; drift 5 mm or less in 10 min at rated load	Not verifiable at TRL 3: worm statically self-locking (back-drive efficiency -0.22) and brake margin 8.2 times, but drift needs a test	Hold test
R9	Hold-to-run	Motion only while the dead-man grip is held and a direction is selected; release stops the clusters within 0.2 s	Not verifiable at TRL 3 (circuit concept only)	Bench test
R10	Flat rolling	Push force 40 N or less at rated load on a smooth floor at walking pace, with the clusters parked	Met: 25 N	Pull test

ID	Requirement	Target	Status at TRL 3 (SCM-CAL-001 v0.2)	Verification (later)
R11	Size	Overall width 600 mm or less; upright height 1,500 mm or less with a fixed handle. Height relaxed from 1,300 mm by Amish, 2026-09-25	Met: 569 mm wide, 1,451 mm tall (model)	Model check; measurement
R12	Affordable	Parts cost \$650 or less for one prototype, charger included. Budget revised from \$600 by Amish, 2026-09-25 (SCM-DDR-002)	Met: \$633 (2.6 % margin)	Priced BOM (bom/ bom.csv)
R13	Battery	24 V class LiFePO4 with BMS and fused output; charging blocked below 0 °C and above 45 °C; full charge in 4 h or less	Met: 3.5 h; protections by specification	Datasheets
R14	Environment	Operate from 0 to 40 °C; electronics IP54 or better; survive rain on outdoor stoops	Not verifiable at TRL 3 (enclosure concept only)	Spray test
R15	Stair and building protection	No steel contact with stairs or walls in normal use; non-marking wheels; skids over stair nosings	Met: the shaft-line guard sweeps 53.9 mm against the 54.2 mm that 32 mm nosings leave free; the countershaft guard clears nosings by 96 mm	Model check; trial climbs

■ Requirements not met or at risk

- **R6 is not met** by 11 % at the plus or minus 6 degree window edge. The longer 150 mm arm widens the swing of the load about its support wheel, so the tighter window decided on 2026-09-25 is not enough on its own. Options (relax R6 to 115 N, tighten the window to plus or minus 4 degrees, or lengthen the handle) are proposed, awaiting Amish (SCM-DDR-002 item 14).
- **R15 is met with no spare radius:** the shaft-line guard is sized to its envelope. A bent guard or a stair outside the R2 range could still be struck.
- **R3, R5 and R12 have thin margins** (5 %, 0.7 kg and 2.6 %) against their revised targets.
- R7, R8, R9 and R14 depend on control, brake and enclosure behavior that needs hardware to verify.

■ Assumptions

- Payload 60 kg; truck 26.3 kg (mass roll-up in SCM-CAL-001); total 86.3 kg.
- The clusters carry the whole weight while lifting (conservative; the operator's grip carries some in practice).
- Center of mass about 500 mm above the cluster shaft; frame tilted about 30 degrees back while climbing.
- The courier stands uphill and pushes or pulls the grip horizontally while climbing.
- Drive efficiencies: motor driver 95 %, brushed motor 80 %, self-locking worm 45 %, first chain stage and bearings 95 %, second chain stage 97 %.
- Rolling resistance coefficient about 0.03 for solid rubber wheels on a smooth floor.
- Stair geometry from the US International Residential Code (196 mm maximum riser, 254 mm minimum tread); national and local codes vary.